

	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 16-01-2024	
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
December 2023



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wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1670	811	59
	4.2 < ws < 7.6	873	1161	532
	ws > 7.6	119	537	2407
Number of points 8169 ; PC=64.1%; EBD=2.2%; POD1=62.7%; POD2=46.3%; POD3=80.3%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2487	216	0
	4.2 < ws < 7.6	175	2153	111
	ws > 7.6	0	140	2887
Number of points 8169 ; PC=92.1%; EBD=0.0%; POD1=93.4%; POD2=85.8%; POD3=96.3%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.1 Title for Contingency tables of Night 1

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wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	7199	91
	12.0 < ws < 18.0	50	718
	ws > 18.0	0	6
Number of points 8169 ; PC=98.1%; EBD=0.0%; POD1=99.3%; POD2=88.1%; POD3=88.6%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

wind direction degree		Observations			
		North	East	South	West
Model data	North	4169	210	14	65
	East	251	347	89	0
	South	38	101	814	80
	West	148	55	48	122

Number of points 6551 ; PC=83.2%; EBD=1.6%; POD1=90.5%; POD2=48.7%; POD3=84.4%; POD4=45.7%

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		Observations			
		North	East	South	West
Model data	North	4554	71	0	8
	East	44	617	13	0
	South	0	25	948	3
	West	8	0	4	256

Number of points 6551 ; PC=97.3%; EBD=0.0%; POD1=98.9%; POD2=86.5%; POD3=98.2%; POD4=95.9%

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

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relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1876	593
	7.1 < rh < 15.3	1352	2215
	rh > 15.3	130	604
Number of points 8052 ; PC=62.8%; EBD=2.0%; POD1=55.9%; POD2=64.9%; POD3=75.6%			

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3105	149
	7.1 < rh < 15.3	252	3213
	rh > 15.3	1	50
Number of points 8052 ; PC=93.6%; EBD=0.0%; POD1=92.5%; POD2=94.2%; POD3=95.0%			

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	8021	0
	50.0 < rh < 70.0	1	30
	rh > 70.0	0	0
Number of points 8052 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=100.0%; POD3=NaN%			

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2435	376	0
	1.9 < pwv < 3.5	426	2031	548
	pwv > 3.5	8	169	1838
Number of points 7831 ; PC=80.5%; EBD=0.1%; POD1=84.9%; POD2=78.8%; POD3=77.0%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2804	70	0
	1.9 < pwv < 3.5	65	2488	45
	pwv > 3.5	0	18	2341
Number of points 7831 ; PC=97.5%; EBD=0.0%; POD1=97.7%; POD2=96.6%; POD3=98.1%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1054	734	195
	0.6 < see < 0.8	406	613	583
	see > 0.8	108	301	650
Number of points 4644 ; PC=49.9%; EBD=6.5%; POD1=67.2%; POD2=37.2%; POD3=45.5%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1581	279	12
	0.6 < see < 0.8	202	1202	125
	see > 0.8	5	152	1435
Number of points 4644 ; PC=84.5%; EBD=0.3%; POD1=88.4%; POD2=73.6%; POD3=91.3%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1481	0	129
	0.6 < see < 0.8	0	1582	129
	see > 0.8	87	66	1299
Number of points 4644 ; PC=93.9%; EBD=4.7%; POD1=94.5%; POD2=96.0%; POD3=83.4%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1785	1	10
	0.6 < see < 0.8	0	1625	9
	see > 0.8	3	7	1562
Number of points 4644 ; PC=99.6%; EBD=0.3%; POD1=99.8%; POD2=99.5%; POD3=98.8%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

coeherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	997	299	8
	$3.4 < \tau < 5.8$	339	805	355
	$\tau > 5.8$	30	466	1000
Number of points		4299 ; PC=65.2%; EBD=0.9%; POD1=73.0%; POD2=51.3%; POD3=73.4%		

Table 15: night1 - Contingency table for coeherence time (ms) in standard configuration and accuracy 0.0

coeherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1299	101	0
	$3.4 < \tau < 5.8$	67	1347	103
	$\tau > 5.8$	0	122	1260
Number of points		4299 ; PC=90.9%; EBD=0.0%; POD1=95.1%; POD2=85.8%; POD3=92.4%		

Table 16: night1 - Contingency table for coeherence time (ms) processed with AR (1H) and accuracy 0.0

coeherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1269	20	123
	$3.4 < \tau < 5.8$	67	1326	123
	$\tau > 5.8$	30	224	1240
Number of points		4299 ; PC=89.2%; EBD=3.6%; POD1=92.9%; POD2=84.5%; POD3=83.4%		

Table 17: night1 - Contingency table for coeherence time (ms) in standard configuration and accuracy 1.22

coeherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1366	0	4
	$3.4 < \tau < 5.8$	0	1555	4
	$\tau > 5.8$	0	15	1359
Number of points		4299 ; PC=99.6%; EBD=0.1%; POD1=100.0%; POD2=99.0%; POD3=99.4%		

Table 18: night1 - Contingency table for coeherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	998	871
	0.6 < glf < 0.7	380	603
	glf > 0.7	121	110
Number of points 4389 ; PC=40.3%; EBD=15.3%; POD1=66.6%; POD2=38.1%; POD3=12.7%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1340	242
	0.6 < glf < 0.7	157	1137
	glf > 0.7	2	205
Number of points 4389 ; PC=80.3%; EBD=0.3%; POD1=89.4%; POD2=71.8%; POD3=80.3%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1359	227	595
	0.6 < glf < 0.7	19	1346	468
	glf > 0.7	121	11	711
Number of points 4389 ; PC=77.8%; EBD=16.3%; POD1=90.7%; POD2=85.0%; POD3=40.1%				

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1496	14	13
	0.6 < glf < 0.7	1	1564	13
	glf > 0.7	2	6	1293
Number of points 4389 ; PC=99.2%; EBD=0.3%; POD1=99.8%; POD2=98.7%; POD3=98.0%				

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	496	190
	4.2 < ws < 7.6	834	2106
	ws > 7.6	49	720
Number of points 7691 ; PC=68.8%; EBD=0.9%; POD1=36.0%; POD2=69.8%; POD3=81.7%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1193	164
	4.2 < ws < 7.6	186	2682
	ws > 7.6	0	170
Number of points 7691 ; PC=90.7%; EBD=0.0%; POD1=86.5%; POD2=88.9%; POD3=94.1%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.2 Title for Contingency tables of Day 1

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wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	6355	90
	12.0 < ws < 18.0	73	835
	ws > 18.0	0	21
Number of points 7691 ; PC=97.1%; EBD=0.0%; POD1=98.9%; POD2=88.3%; POD3=87.7%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	3070	191	0		80
	East	47	110	22		0
	South	4	11	510		139
	West	1303	10	55		1524

Number of points 7076 ; PC=73.7%; EBD=0.2%; POD1=69.4%; POD2=34.2%; POD3=86.9%; POD4=87.4%

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	4237	35	0		116
	East	24	279	3		0
	South	0	8	569		16
	West	163	0	15		1611

Number of points 7076 ; PC=94.6%; EBD=0.0%; POD1=95.8%; POD2=86.6%; POD3=96.9%; POD4=92.4%

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

FATE - Monthly Report

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	635	323
	7.1 < rh < 15.3	1090	3192
	rh > 15.3	85	723
Number of points 7666 ; PC=67.6%; EBD=1.1%; POD1=35.1%; POD2=75.3%; POD3=83.7%			

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1614	125
	7.1 < rh < 15.3	196	4014
	rh > 15.3	0	99
Number of points 7666 ; PC=93.7%; EBD=0.0%; POD1=89.2%; POD2=94.7%; POD3=96.0%			

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	7660	0
	50.0 < rh < 70.0	6	0
	rh > 70.0	0	0
Number of points 7666 ; PC=99.9%; EBD=0.0%; POD1=99.9%; POD2=NaN%; POD3=NaN%			

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1435	247	3
	1.9 < pwv < 3.5	597	2269	531
	pwv > 3.5	0	226	2142
Number of points 7450 ; PC=78.5%; EBD=0.0%; POD1=70.6%; POD2=82.7%; POD3=80.0%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1957	60	0
	1.9 < pwv < 3.5	75	2650	46
	pwv > 3.5	0	32	2630
Number of points 7450 ; PC=97.1%; EBD=0.0%; POD1=96.3%; POD2=96.6%; POD3=98.3%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)